

Notes: Keys and Wang (JFE, 2019)

Minimum Payments and Debt Paydown in Consumer Credit Cards

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1 What the paper is doing

Credit-card repayment is a central household finance margin because it governs the speed of debt paydown and therefore interest costs. The paper asks whether consumers who pay “at or near the minimum” do so mainly because they are *liquidity constrained* or because the minimum payment is a *salient anchor* that shapes choices even when consumers could pay more. The authors begin with descriptive evidence that minimum and near-minimum payments are common and weakly related to income or age but strongly related to balance and credit score. They then exploit issuer-level changes in minimum-payment formulas as quasi-experiments: under pure liquidity constraints, raising the minimum should cause bunching at the new minimum without changing the fraction of payments below it; under anchoring, raising the minimum shifts the entire distribution upward so that fewer accounts pay at or below the new minimum. Event-study difference-in-differences estimates confirm the anchoring pattern for both minimum-payment increases *and* decreases, implying a sizable anchoring population. Finally, the paper interprets these results through behavioral choice architecture and discusses policy implications for consumer credit regulation.

2 Institutional background: what is a credit-card minimum payment?

2.1 Minimum payment formulas and why they change

In a typical general-purpose credit card, each monthly statement lists a *minimum payment* that keeps the account in good standing. Issuers determine this minimum using a formula that usually contains:

- a **floor** (a flat dollar amount for low balances), and
- a **slope** (a percentage of the balance for higher balances),

often with additional components (e.g., interest, fees, promotional rules).

A stylized example in the paper is:

$$\text{minimum} = \max\{\text{floor}, 0.02 \times \text{balance}\}. \quad (1)$$

Issuers sometimes change formulas (e.g., raise the floor from \$20 to \$40, or shift the whole schedule up) due to business strategy and regulatory pressure (e.g., to ensure balances amortize “over a reasonable period” as suggested by regulators).

2.2 Two stylized formula changes (Figure 1)

Figure 1 illustrates two common changes:

1. **Floor increase:** raise the flat minimum for low balances. This affects only accounts with balances low enough that the floor binds.

2. **Schedule shift:** increase the minimum across a wider set of balances by shifting the schedule upward.

These features matter for identification: whether an account is “treated” by a formula change can depend on its balance and other characteristics.

3 Data and key variables

3.1 Data: CFPB Credit Card Database (CCDB)

The CCDB covers monthly account-level outcomes for a large fraction of the U.S. general-purpose credit card market (2008–2013 in the sample used here), linked to credit bureau information (e.g., FICO at origination, income proxies, total credit portfolio). The paper uses a random sample of active accounts from included issuers and observes, for each account-month: minimum payment due, actual payment, balances, APRs, fees, and delinquency indicators.

3.2 Summary statistics (Table 1) — what to notice

Table 1 shows that borrowers are middle-aged on average (about 52), have average income around \$66k and FICO around 701, and hold multiple cards (about 3 open cards). Average statement balance is about \$3.2k and utilization is about 45%. The repayment distribution is extremely dispersed:

- mean fraction paid is 42% but the median is only 11% (many low payments) and the 75th percentile is 100% (many full payments);
- the average minimum payment is about \$82 while the average actual payment is about \$570.

The key decomposition of payment behavior (by account-month) is:

less than minimum (9%), exact minimum (15%), near minimum (20%), intermediate (23%), full (33%).

Thus, about **44%** of account-months are at or within \$50 above the minimum payment (the “minimum/near-minimum region”).

Table 1
Summary statistics.
The table provides summary statistics from our sample. The sample consists of a 1% sample of active general-purpose credit card accounts from several large issuers from 2008–2013, with approximately 40 million monthly account-level observations.

	Mean	Std. dev.	p25	p50	p75
Card and account					
Income	\$65,583	\$49,300	\$32,400	\$55,000	\$87,499
FICO at origination	701	76	656	707	758
Account age (years)	9.14	7.90	3	7	14
Age of primary account-holder	51.68	16.10	39	51	63
Credit limit	\$9,767	\$8,842	\$3,000	\$8,100	\$13,760
Retail APR	16.14	8.10	12	15.24	19.99
Joint account	8%				
Has annual fee	12%				
All card accounts					
# of Open cards	3.02	2.30	1	2	4
Total balance	\$10,677	\$16,859	\$1,227	\$4,541	\$13,751
Total credit limit	\$38,928	\$35,779	\$14,500	\$29,600	\$52,900
# of New cards in last 3 months	0.06	0	0	0	0
# of Cards 60+ days past due	0.03	0	0	0	0

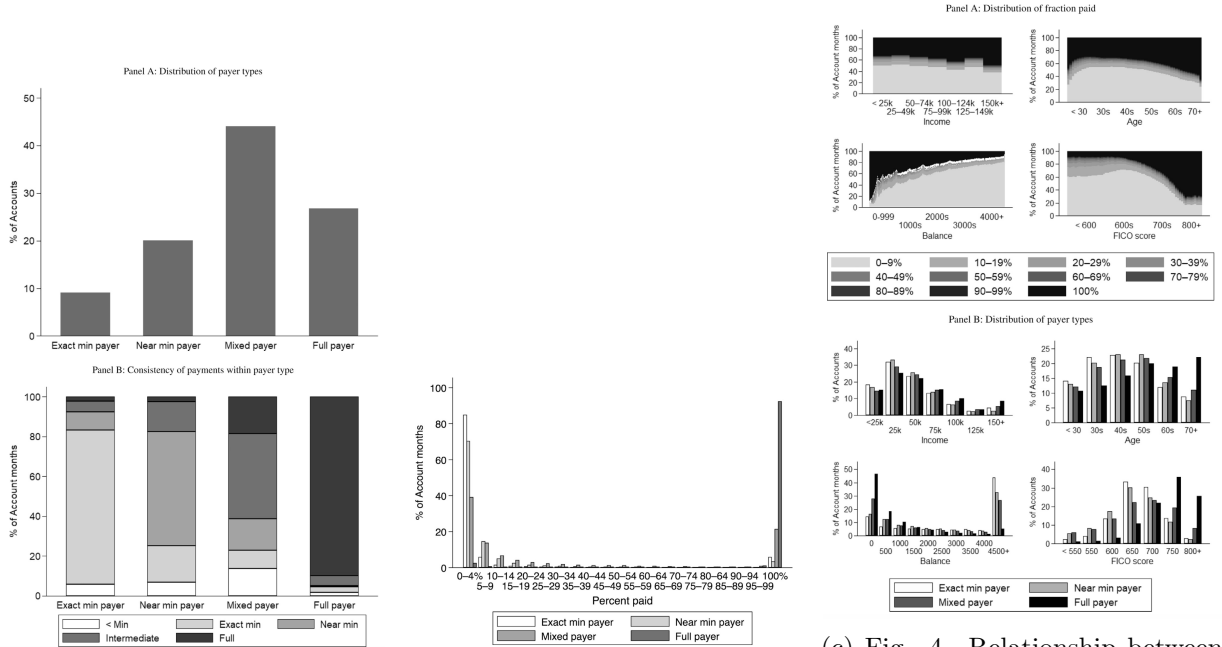
Purchases and balances					
Utilization	45%	40%	6%	35%	87%
Balance	\$3,187	\$4,607	\$384	\$1,359	\$4,187
Promotional	\$545	\$2,129	\$0	\$0	\$0
Cash advance	\$168	\$988	\$0	\$0	\$0
Penalty	\$398	\$1,940	\$0	\$0	\$0
Purchase volume	\$501	\$1,497	\$0	\$62	\$445
Purchase volume > 0	63%				
Payment and delinquency behavior					
Fraction paid	42%	50%	3%	11%	100%
Minimum payment	\$82	\$210	\$20	\$39	\$88
Actual payment	\$570	\$1,651	\$50	\$150	\$451
Payment:					
< Minimum	9%				
Exact minimum	15%				
Near minimum	20%				
Intermediate	23%				
Full	33%				
Charged fees:					
Late	9%				
Overlimit	1%				
NSF	0.2%				
Has past due	8%				

4 Descriptive evidence: what repayment patterns look like

4.1 Payer-type classification (Figures 2–4)

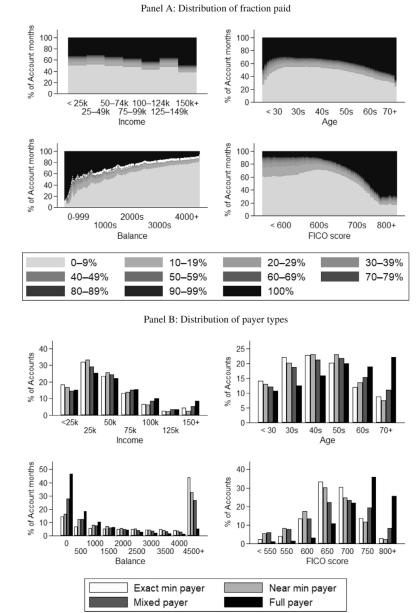
The paper classifies accounts into **payer types** based on whether they pay in full or at/near the minimum in at least 50% of months:

- **Exact min payer:** pays exactly the minimum in $\geq 50\%$ of months.
- **Near min payer:** pays strictly more than the minimum but within \$50 above it in $\geq 50\%$ of months.
- **Full payer:** pays in full in $\geq 50\%$ of months.
- **Mixed payer:** all remaining accounts (none of the above behaviors dominates).



(a) Fig. 2. Prevalence of minimum and near-minimum payments.

(b) Fig. 3. Distribution of payments as a fraction of balance.



(c) Fig. 4. Relationship between payments and borrower characteristics.

Figure 2 (prevalence + within-type composition). Panel A shows that “mixed” accounts are the largest group, but *near-minimum* accounts are much more common than *exact-minimum* accounts. Panel B confirms that the classification aligns with behavior: exact-minimum accounts overwhelmingly make exact-min payments; near-min accounts concentrate mass in the near-min region; full payers overwhelmingly pay in full.

Figure 3 (payments as percent of balance). Figure 3 shows a bimodal pattern:

- a large spike at very low percentages (roughly 0–4% of balance), corresponding to minimum/near-minimum paying,

- and a large spike at 100%, corresponding to full payment.

This illustrates why repayment is best thought of as a choice among discrete “heuristic” regions rather than a smooth interior optimization.

Figure 4 (payments vs borrower/account characteristics). Panel A shows how the distribution of fraction paid changes with income, age, balance, and FICO. Panel B shows how payer types vary with those characteristics. The central descriptive pattern emphasized in the text is:

- **balance and FICO** are strongly correlated with paying near the minimum (higher balances and lower FICO $\Rightarrow$ more low payments),
- **income and age** are surprisingly weak correlates, which already suggests liquidity constraints alone may be insufficient.

4.2 Rounding and “rules of thumb” (Figure 5)

Figure 5 zooms in on near-minimum payments. Panel A plots dollars paid relative to the minimum and shows visible spikes at round numbers above the minimum. Panel B normalizes by a *rounded* minimum (minimum rounded up to the next \$25) and shows large spikes at \$0 and other multiples, indicating that many consumers appear to start from the minimum and then apply a rounding heuristic (e.g., “pay the minimum rounded up to \$25/\$50/\$100”). This motivates an anchoring interpretation: the minimum payment acts as a salient starting point.

5 Conceptual framework: liquidity constraints vs anchoring

5.1 Setup: a formula change from Min_1 to Min_2

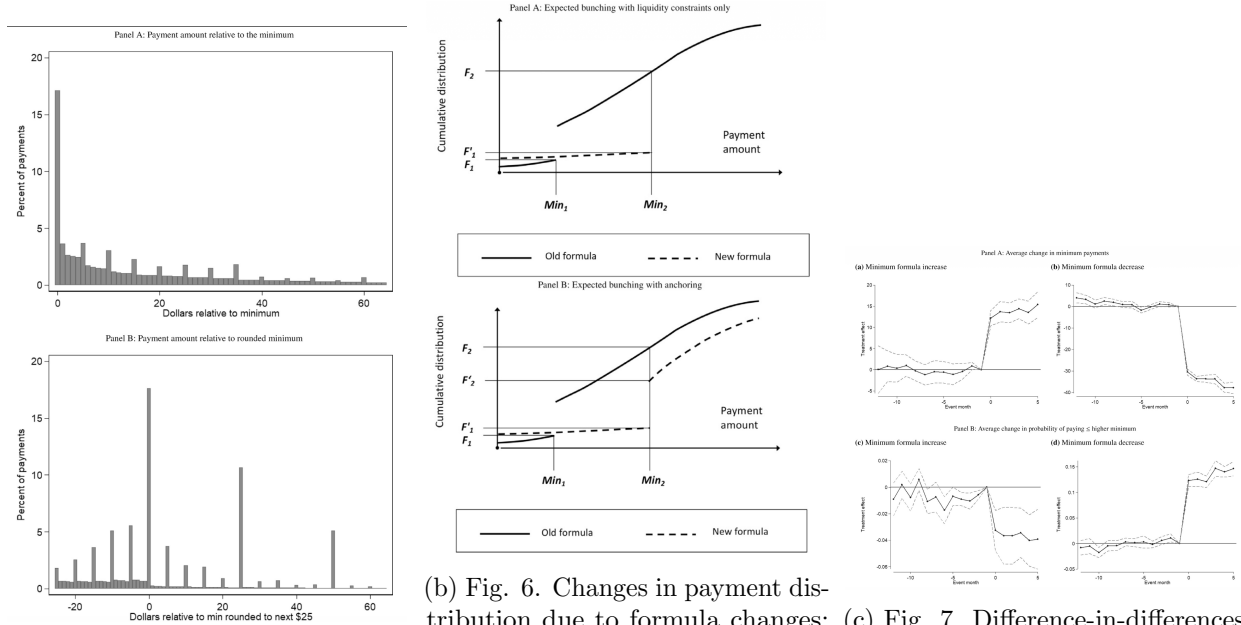
Consider an issuer that changes its formula so that an account’s minimum payment increases from Min_1 (old formula) to Min_2 (new formula), with $Min_2 > Min_1$ (the decrease case is analogous).

Let F_1 be the fraction delinquent under the old formula (severe liquidity constraint), and F'_1 the fraction delinquent under the new formula. Let F_2 be the fraction of consumers who pay *less than or equal to* Min_2 under the old formula, and F'_2 the corresponding fraction under the new formula.

5.2 Predictions (Figure 6)

Figure 6 presents two competing predictions for what happens when Min_1 rises to Min_2 .

Liquidity constraints only (Panel A). Accounts paying between Min_1 and Min_2 under the old formula will tend to *bunch mechanically* at Min_2 under the new formula (or become delinquent if they cannot meet Min_2). In this case, the mass of payments *at or below* Min_2 should not fall: intuitively, consumers remain in the “ $\leq Min_2$ ” region even after the hike.



(a) Fig. 5. Distribution of near-minimum payments.

(b) Fig. 6. Changes in payment distribution due to formula changes: Liquid- ity constraints vs. anchoring.

(c) Fig. 7. Difference-in-differences regressions around issuer formula changes.

Anchoring (Panel B). If some consumers choose an interior payment amount by starting at the minimum and adjusting upward (anchoring-and-adjustment), then increasing the minimum shifts the entire distribution upward, and some consumers who used to pay between Min_1 and Min_2 will now pay *strictly more than* Min_2 . In this case, the fraction paying at or below Min_2 declines: $F'_2 < F_2$.

5.3 A key estimand: change in the share paying $\leq Min_2$

Define

$$\beta \equiv F'_2 - F_2.$$

Under anchoring and $Min_2 > Min_1$, we expect $\beta < 0$. If θ denotes the fraction of *affected accounts* that anchor (among those who used to be in the “ $\leq Min_2$ ” region), then the paper argues that

$$-\beta/F_2$$

provides a **lower bound** on θ (because some anchoring movements are not captured by this particular test).

5.4 From affected accounts to the population: lower bound on overall anchoring (Equation (2))

The paper imposes additional structure: anchoring accounts do not adjust beyond a fixed bandwidth above the minimum. Let the bandwidth be X (baseline $X = \$50$) and define

$$\delta \equiv CDF(Min_1 + X),$$

the share of payments within \$50 above the minimum (including below-min and exact-min payments).

Then a lower bound on the fraction of *all accounts* that anchor to the minimum payment is:

$$\theta^* \geq -\beta \delta / F_2. \quad (2)$$

This formula is central to how Table 2 converts regression estimates into economically interpretable prevalence numbers.

5.5 Upper bound on liquidity-constrained prevalence

Within the same logic, the fraction of accounts paying close to the minimum *but not classified as anchoring by the test* can be treated as an upper bound on “liquidity constraints” in the near-minimum region:

$$LC^* \equiv \delta(1 - \theta),$$

which is reported as a benchmark in Table 2 and Table 3.

6 Identification strategy and empirical models (all equations)

6.1 Event-study difference-in-differences around issuer formula changes

The identification strategy compares accounts *affected* by a formula change to accounts *not affected* by that same issuer change and to accounts at issuers with no change, before vs after the event.

6.2 Model 1: treatment indicator for exposure (Equation (3))

Let Min_{ijt}^1 denote account i ’s minimum payment under issuer j ’s *old* formula in month t , and Min_{ijt}^2 under the *new* formula. Define:

$$I_{ijt} = \begin{cases} 1 & \text{if } Min_{ijt}^1 \neq Min_{ijt}^2, \\ 0 & \text{if } Min_{ijt}^1 = Min_{ijt}^2. \end{cases} \quad (3)$$

This captures whether a given account-month is “treated” by the formula change (treatment varies by balance and other features).

6.3 Model 2: baseline event-study DD regression (Equation (4))

Let r_{jt} be event time (months since issuer j implemented the formula change). Estimate:

$$Y_{ijt} = \alpha_i + \eta_t + \sum_{\tau} \beta_{\tau} (I_{ijt} \times \mathbf{1}\{r_{jt} = \tau\}) + \gamma X_{ijt} + \varepsilon_{ijt}, \quad (4)$$

where:

- Y_{ijt} is an outcome (minimum due, delinquency, or a payment-distribution indicator),

- α_i are account fixed effects (in the microdata version),
- η_t are month fixed effects,
- X_{ijt} are controls (balance deciles, credit limit, FICO, purchases, APR, promotional flags, etc., depending on specification),
- the coefficients of interest are the event-time effects β_τ , normalized by setting the pre-period coefficient to zero (e.g., $\beta_{-1} = 0$).

Standard errors are clustered conservatively at the interaction of FICO decile and issuer formula type.

6.4 Model 3: indicator for paying at/below the higher minimum (Equation (5))

To operationalize F_2 and F'_2 , define:

$$P_{ijt} \equiv \mathbf{1}(\text{payment}_{ijt} \leq \max\{\text{Min}_{ijt}^1, \text{Min}_{ijt}^2\}). \quad (5)$$

For a minimum *increase*, $\max\{\text{Min}^1, \text{Min}^2\} = \text{Min}^2$. If anchoring is present, some accounts move from paying $\leq \text{Min}^2$ to paying $> \text{Min}^2$, reducing P .

6.5 Model 4: intensity specification using dollar changes (Equations (6)–(7))

Define the account-month change in minimum payment implied by the formula change:

$$\Delta \text{Min}_{ijt} \equiv \text{Min}_{ijt}^2 - \text{Min}_{ijt}^1. \quad (6)$$

An alternative event-study uses intensity:

$$Y_{ijt} = \alpha_i + \eta_t + \sum_{\tau} \beta_{\tau} (\Delta \text{Min}_{ijt} \times \mathbf{1}\{r_{jt} = \tau\}) + \gamma X_{ijt} + \varepsilon_{ijt}. \quad (7)$$

This checks robustness to using the magnitude of the formula-induced change (rather than the binary exposure).

7 Main causal results: interpreting Table 2 and Figure 7

7.1 Figure 7: no pre-trends and immediate, persistent effects

Figure 7 plots the estimated β_τ coefficients around the event month.

- Panel A shows a sharp jump in the average minimum payment when the formula changes (upward for increases, downward for decreases), with no systematic pre-trends and persistent post effects. This validates the “first stage” of the design.
- Panel B shows the corresponding change in P (probability of paying at or below the higher minimum). For increases, P falls after the event; for decreases, P rises after the event. Both patterns match anchoring-and-adjustment.

7.2 Table 2: mapping regression outputs to anchoring prevalence

Table 2 reports three types of objects:

1. **Regression outcomes:** change in minimum due (first stage), delinquency, and P .
2. **Anchoring prevalence:** θ and θ^* computed using the conceptual framework.
3. **Liquidity-constraint upper bound:** LC^* for context.

Columns (1)–(3): regression estimates. For positive formula changes (Panel B, preferred “full controls”):

- the minimum due rises by about \$12–\$15;
- delinquency barely moves and is statistically weak;
- P declines by about 3–4 percentage points within 3–5 months.

Under pure liquidity constraints, P should not fall materially (it might even rise via delinquency). Thus, the negative P effect is evidence of anchoring.

For the one negative formula change (Panel C): the minimum due drops by about \$30–\$38 and P *increases* by 12–15 percentage points. This is especially powerful evidence for anchoring because the *incentives to pay more than the old minimum* are unchanged when the minimum falls; yet many consumers reduce their payment in response to the lower salient minimum.

Columns (4)–(6): anchoring and liquidity interpretation. Combining the conceptual mapping, the paper concludes:

- **Positive formula changes:** at least about 18–22% of affected accounts anchor (θ), implying roughly 8–9% of all accounts anchor (θ^*).
- **Negative formula change:** anchoring prevalence is even larger, with θ around 31–38% and θ^* around 16–20%.
- LC^* is around one-third in the preferred positive-change specification, suggesting that liquidity constraints are present but cannot explain the entire near-minimum region once anchoring is accounted for.

7.3 Economic interpretation (why these magnitudes matter)

Anchoring means that a salient contractual term can shift repayment behavior even when it is not a binding constraint. Because revolving interest rates are high, paying closer to the minimum than is optimal can substantially slow debt paydown and raise lifetime borrowing costs. Thus, repayment choices are shaped not only by constraints and preferences but also by *choice architecture* embedded in monthly statements.

Table 2

Estimates of anchoring from minimum payment formula changes. The table shows difference-in-differences regression estimates for the effect of issuer formula changes on payments. Panels A and B report pooled estimates for four formula changes that increased the minimum payment, and Panel C reports estimates for one formula change that decreased the minimum payment. Standard errors clustered by issuer formula type interacted with FICO decile are shown in parentheses, and p -values are shown in brackets. Columns 1 and 2 show the average dollar change in minimum payments and the share of delinquent accounts, respectively. Column 3 presents the change in P , the share of accounts paying less than or equal to the higher of the two minimum payment formulas. Columns 4 and 5 present estimates of θ and θ^* , the fraction of anchoring accounts among those directly affected by the formula changes and among all accounts, respectively. Column 6 presents an upper-bound estimate of the share of accounts that pay close to the minimum payment due to liquidity constraints. See Section 4.1 for details. The regressions in Panel A include only time fixed effects and fixed effects for issuer formula type interacted with FICO decile as independent variables, and includes only potentially treated accounts in the analysis sample. Panels B and C include all accounts from control and treated issuers and a full set of account-level controls and fixed effects for issuer formula type interacted with FICO decile.

	Regression estimates			Anchoring estimates		
	Minimum due (1)	Delinq. (2)	P (3)	θ (4)	θ^* (5)	LC^* (6)
Panel A: Positive formula changes, no time-varying controls						
$\Delta (t = -1 \text{ to } t = 0)$	14.0 (0.7) [0.0]	0.016 (0.002) [0.000]	-0.013 (0.007) [0.056]	0.07 (0.04)	0.03 (0.02)	0.39 (0.02)
$\Delta (t = -1 \text{ to } t = 3)$	14.3 (0.6) [0.0]	0.002 (0.003) [0.380]	-0.031 (0.006) [0.000]	0.17 (0.04)	0.07 (0.01)	0.35 (0.01)
$\Delta (t = -1 \text{ to } t = 5)$	14.5 (0.7) [0.0]	0.002 (0.003) [0.400]	-0.037 (0.008) [0.000]	0.21 (0.04)	0.09 (0.02)	0.33 (0.02)
Panel B: Positive formula changes, full controls						
$\Delta (t = -1 \text{ to } t = 0)$	12.0 (2.0) [0.0]	0.004 (0.003) [0.210]	-0.003 (0.008) [0.000]	0.18 (0.04)	0.08 (0.02)	0.34 (0.02)
$\Delta (t = -1 \text{ to } t = 3)$	14.4 (1.2) [0.0]	0.000 (0.002) [0.900]	-0.034 (0.009) [0.000]	0.19 (0.05)	0.08 (0.02)	0.34 (0.02)
$\Delta (t = -1 \text{ to } t = 5)$	15.4 (1.6) [0.0]	0.002 (0.004) [0.580]	-0.039 (0.011) [0.001]	0.22 (0.06)	0.09 (0.04)	0.33 (0.03)
Panel C: Negative formula change, full controls						
$\Delta (t = -1 \text{ to } t = 0)$	-30.5 (0.7) [0.0]	0.005 (0.003) [0.071]	0.120 (0.006) [0.000]	0.31 (0.02)	0.16 (0.01)	0.50 (0.01)
$\Delta (t = -1 \text{ to } t = 3)$	-33.7 (1.1) [0.0]	0.007 (0.002) [0.003]	0.150 (0.008) [0.000]	0.38 (0.02)	0.20 (0.01)	0.31 (0.01)
$\Delta (t = -1 \text{ to } t = 5)$	-37.8 (1.4) [0.0]	0.002 (0.003) [0.390]	0.150 (0.007) [0.000]	0.38 (0.02)	0.20 (0.01)	0.31 (0.01)

Table 3

Heterogeneity in anchoring estimates.

The table shows heterogeneity in estimates of anchoring from difference-in-differences regressions stratified by borrower characteristics. The first column in each group presents the estimate for the change in P , the share of accounts paying less than or equal to the higher of the two minimum payment formulas. The second column presents estimates of θ^* , the fraction of all accounts that anchor to the minimum payment. The third column presents estimates of the share of accounts that pay close to the minimum payment due to liquidity constraints. Standard errors clustered by issuer formula type interacted with FICO decile are shown in parentheses, and p -values are shown in brackets. Panel A stratifies the results by FICO at origination, Panel B stratifies by payer type, Panel C stratifies by income, and Panel D stratifies by borrower age. All regressions include both treated and control issuers and controls for time-varying account characteristics and fixed effects for issuer formula type interacted with FICO decile.

	P^* (1)	θ^* (2)	LC^* (3)	P^* (4)	θ^* (5)	LC^* (6)	P^* (7)	θ^* (8)	LC^* (9)	P^* (10)	θ^* (11)	LC^* (12)
Panel A: Stratified by FICO band												
Sample:	Deep subprime < 620			Core subprime 620-650			Prime 660-720			Super-prime > 720		
$\Delta (t = -1 \text{ to } t = 5)$	-0.072 (0.008) [0.000]	0.13 (0.01) [0.001]	0.59 (0.01) [0.000]	-0.080 (0.010) [0.000]	0.15 (0.02) [0.000]	0.52 (0.02) [0.000]	-0.066 (0.015) [0.000]	0.15 (0.03) [0.000]	0.38 (0.03) [0.000]	-0.020 (0.010) [0.049]	0.06 (0.03) [0.000]	0.15 (0.03) [0.000]
Panel B: Stratified by payer type												
Sample:	Exact min payer			Near min payer			Mixed payer			Full payer		
$\Delta (t = -1 \text{ to } t = 5)$	-0.097 (0.013) [0.000]	0.11 (0.01) [0.001]	0.83 (0.01) [0.000]	-0.160 (0.016) [0.000]	0.32 (0.03) [0.000]	0.53 (0.03) [0.000]	-0.036 (0.007) [0.000]	0.08 (0.02) [0.000]	0.21 (0.02) [0.000]	0.004 (0.002) [0.048]	0.00 (0.00) [0.000]	0.02 (0.00) [0.000]
Panel C: Stratified by income quartile												
Sample:	Lowest			Second			Third			Highest		
$\Delta (t = -1 \text{ to } t = 5)$	0.046 (0.013) [0.000]	0.10 (0.03) [0.000]	0.36 (0.03) [0.000]	-0.045 (0.016) [0.001]	0.10 (0.03) [0.001]	0.36 (0.03) [0.000]	-0.036 (0.012) [0.005]	0.08 (0.03) [0.000]	0.32 (0.03) [0.000]	-0.033 (0.011) [0.004]	0.07 (0.02) [0.000]	0.29 (0.02) [0.000]
Panel D: Stratified by age quartile												
Sample:	Lowest			Second			Third			Highest		
$\Delta (t = -1 \text{ to } t = 5)$	-0.064 (0.015) [0.000]	0.13 (0.03) [0.000]	0.35 (0.03) [0.000]	-0.045 (0.100) [0.000]	0.10 (0.02) [0.000]	0.39 (0.02) [0.000]	-0.041 (0.012) [0.000]	0.11 (0.03) [0.000]	0.31 (0.03) [0.000]	-0.034 (0.009) [0.009]	0.08 (0.03) [0.000]	0.24 (0.03) [0.000]

8 Heterogeneity: interpreting Table 3 (who anchors?)

Table 3 stratifies the design and recomputes P , θ , and LC^* for different groups.

8.1 Panel A: by credit score (FICO bands)

Anchoring θ is *remarkably similar* for most of the credit-score distribution (deep subprime, core subprime, prime) and only clearly lower for super-prime. In contrast, the liquidity upper bound LC^* declines monotonically with FICO, consistent with better credit being associated with fewer binding liquidity constraints. This separation supports the paper’s central claim: drivers of anchoring differ from drivers of credit risk.

8.2 Panel B: by payer type

Near-minimum payers are the most likely to anchor (largest θ), which aligns with the core behavioral hypothesis. Full payers essentially do not anchor (near-zero θ) and have very low LC^* . Exact-minimum payers show very high LC^* , consistent with a large constrained component in the exact-min group.

8.3 Panels C–D: by income and age quartiles

Anchoring varies only modestly with income and age, whereas liquidity constraints fall with higher income and older age. This pattern directly undermines the simple “low payments = liquidity

constrained” narrative and supports the behavioral anchoring channel.

9 Conclusion

9.1 Summary

This paper concludes that **minimum payments are not only a contractual constraint but also a salient behavioral anchor that causally shapes repayment choices**. Using near-universe U.S. credit-card account data and quasi-experimental **issuer-level changes in minimum-payment formulas**, the authors show that many borrowers adjust their actual payments *in the direction of the new minimum*, even when the change does not operate through standard incentive channels in a frictionless optimizing model.

The empirical story is internally consistent and proceeds in four steps:

- (i) **Minimum and near-minimum payments are prevalent.** In the raw data, a large share of account-months clusters at the minimum or close to it: about 9% pay less than the minimum, 15% pay exactly the minimum, and 20% pay within \$50 above it (Table 1). This is too common to treat as a fringe behavior.
- (ii) **The minimum payment behaves like a reference point.** Payment distributions show discrete mass points at the minimum and at rounded increments above it (e.g., “minimum rounded up to \$25/\$50”), consistent with anchoring-and-adjustment rules of thumb rather than smooth interior optimization.
- (iii) **Quasi-experimental formula changes identify a causal anchoring response.** The key test compares two competing predictions when the issuer changes the minimum formula: under pure liquidity constraints, raising the minimum should mechanically shift some payments up to the new minimum but should not reduce the fraction paying at/below the new (higher) minimum; under anchoring, raising the minimum shifts the distribution upward so that more accounts pay *above* the new minimum (and the at/below share falls).
- (iv) **The data support anchoring—for both increases and decreases.** For the four positive formula changes (preferred specification with full controls, $t = -1$ to $t = 5$), the minimum due rises by about \$15 and the share paying at or below the higher minimum falls by about 3.9 percentage points (Table 2, Panel B). For the one negative formula change, the minimum due falls by about \$38 and the at/below share *increases* by about 15 percentage points (Table 2, Panel C). The fact that payments move down when the minimum is lowered is particularly strong evidence for anchoring, because “paying less” is not mechanically required.

9.2 Key quantitative conclusions

The paper converts the distributional response into interpretable prevalence measures (lower bounds):

- **Positive formula changes:** at least $\theta \approx 0.22$ of directly affected accounts anchor, implying at least $\theta^* \approx 0.09$ of *all* accounts anchor to the minimum payment (Table 2, Panel B, $t = -1$ to $t = 5$).
- **Negative formula change:** anchoring is even larger: $\theta \approx 0.38$ among affected accounts and $\theta^* \approx 0.20$ among all accounts (Table 2, Panel C, $t = -1$ to $t = 5$).
- **Liquidity constraints still matter, but do not explain the whole near-minimum region.** The paper reports an upper-bound share of accounts paying close to the minimum due to liquidity constraints of about $LC^* \approx 0.33$ for the positive changes and $LC^* \approx 0.31$ for the negative change (Table 2). Hence, a substantial fraction of minimum/near-minimum payments is not well explained by constraints alone.
- **Heterogeneity reinforces the behavioral interpretation.** Anchoring is much higher among *near-minimum payers* (lower bound $\theta^* \approx 0.32$) and essentially zero among *full payers* (Table 3, Panel B). Across credit-score groups, anchoring is similar for most borrowers (roughly 13–15%) but smaller among super-prime borrowers (about 6%), while liquidity-constraint measures decline monotonically with credit score (Table 3, Panel A).

9.3 Why minimum payments change behavior

The paper’s intuition is that repayment decisions often follow a **two-stage heuristic** rather than a fully optimized intertemporal plan:

Step 1: the statement provides a salient default-like reference point. The minimum payment is printed prominently, has a clear “must pay to stay current” interpretation, and is naturally interpreted by many borrowers as a recommendation or benchmark.

Step 2: borrowers adjust imperfectly around that reference point. Many borrowers appear to start from the minimum and then adjust upward using simple rules (e.g., rounding to a nearby convenient number). This produces bunching at the minimum and at rounded increments above it.

Step 3: when the minimum changes, the anchor moves. If repayment is anchored, then changing the formula shifts the entire choice set *as perceived by the borrower*: payments drift up when the minimum rises and drift down when it falls. This behavior can occur even when standard incentive channels are weak, which is why the observed sign patterns (especially for the minimum decrease) are informative.

Step 4: constraints and anchoring coexist. Some borrowers are genuinely constrained and will pay the minimum because they cannot pay more. But the paper shows that constraints alone cannot explain the aggregate mass near the minimum, nor the direction of the distributional shifts after formula changes.

9.4 Implications (policy and research)

The paper implies that **choice architecture in consumer credit markets can materially affect debt dynamics**. In practice:

- Raising minimum payments may accelerate paydown for an anchoring population without a large increase in delinquency (Table 2 shows delinquency changes are small).
- Conversely, lowering the minimum can slow paydown by pulling anchored borrowers downward.
- Disclosure design (e.g., highlighting alternative “suggested payments” that amortize balances faster) can be a policy lever comparable in importance to changing APRs or fees when borrowers follow heuristics.

9.5 Takeaway

Issuer-driven minimum-payment formula changes reveal that a sizable fraction of credit-card borrowers anchor their monthly payments to the minimum payment cue—moving payments up when the minimum rises and down when it falls—so minimum payments shape debt paydown not only through constraints but also through behavioral salience.